

Sean Yoo

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EDUCATION

New York University, Courant Institute

Bachelor of Arts, Data Science; Minor, History

Sep 2023 - May 2027

New York, NY

- **GPA:** 3.5
- **Coursework:** Machine Learning, Databases for Business Analytics, Probability & Statistics, Linear Algebra, Calculus II, Data Structures, Physics

SKILLS

- **Technical:** Python, SQL, Java, Git, BigQuery, Spyder, JavaScript, MS Suite, GitHub, Supervised & Unsupervised Learning, Deep Learning, Tableau
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, Statsmodels, SciPy, NLTK, yfinance, Chart.js, React

EXPERIENCE

HCMpoint

Apr 2026 - Present

Web Development Intern

New York, NY

- Engineered a full-stack **trade show lead capture system** for a conference booth that generated **15+ qualified leads**, combining a branded HTML/CSS registration form, animated thank-you page, and Google Apps Script backend
- Built a **Google Sheets integration** via Apps Script to automatically log attendee submissions in real time with timestamped rows mapped to dynamic column headers
- Implemented **automated email confirmation** on form submission using MailApp, delivering personalized HTML emails to attendees
- Added **security and abuse-prevention** measures including token-based authentication, a honeypot field, and per-email rate limiting via Apps Script's Cache Service

NYU Academic Advising Center

May 2025 - Present

Peer Advising Leader

New York, NY

- Developed a **full-stack degree planning web application** for NYU advisors using React and JavaScript with support for **1,800+ curriculum records**. An algorithm that generates semester-by-semester graduation plans based on prereqs, and credit requirements
- Optimized transfer credit data preparation for academic departments by building Python scripts to **streamline institution/CEEB code lookup**, eliminating hours of manual processing per intake cycle
- Served as the primary point of contact for students across in-person, phone, and email channels, leveraging Microsoft Office, Google Suite, Tableau, and NYU Connect to resolve advising inquiries

NYU Stern

Aug 2024 - Jan 2025

DART Data Operations Analyst

- Automated data cleaning and deduplication for **2,400+ alumni records** using Python and Excel, reducing query time by **~30%** and improving outreach reliability for the development team
- Maintained constituent data (addresses, employment records, engagement logs) in **Blackbaud CRM**, ensuring accuracy for alumni communications across the Stern development office
- Monitored and processed returned mail logs to proactively **flag data quality issues** and uphold contact database integrity

PROJECTS

Comparing Logistic Regression and XGBoost on Spotify Audio Features

Apr 2026 - May 2026

Principles of Data Science II Capstone

- Built a **music classification pipeline** in Python for **50,000 Spotify tracks** across 10 genres, achieving a peak **0.9306 AUC** with XGBoost over Logistic Regression
- Applied a **structural cluster feature** using **UMAP and K-Means** that became the top predictor by XGBoost feature importance
- Conducted **Pearson correlation analysis** on 11 audio features, mapping a loudness-energy-acousticness axis that isolated distinct genre separability patterns
- Evaluated performance via **confusion matrices** and **PR/ROC curves**, diagnosing clear classification failure modes
- Prevented **data leakage** by isolating StandardScaler transformations to the training set within a **stratified 90/10 split**

Reinforcement Learning Agent for Stock Prediction

May 2026 - Present

Personal Project

- Engineered a **Q-Learning trading agent** in Python that discretizes RSI, Bollinger Bands, and MACD into a finite state space for automated trading signals
- Designed a **Sharpe-inspired reward function** combining directional PnL with a volatility penalty to maximize risk-adjusted returns
- Implemented a **walk-forward training loop** over 150 episodes with epsilon decay from 1.0 to 0.01 across full price histories
- Applied the **Bellman equation** for Q-table updates, tracking per-episode cumulative returns and annualized Sharpe ratios to verify convergence